

Lesson 5.1 – Cutting at the Branch Collar

Definition

The branch collar is the swollen area where a branch connects to the trunk or a larger branch.

Importance

- Promotes natural healing
 - Prevents disease entry
 - Protects the tree structure
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Correct Method

- Cut just outside the branch collar
 - Make a clean, smooth cut
 - Follow the natural angle of the branch
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What to Avoid

- Cutting flush against the trunk
 - Leaving long stubs
 - Making rough or jagged cuts
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Example

Cutting a mango branch:

- Correct cut → heals quickly without damage
 - Incorrect cut → leads to decay or infection
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Demo (Step-by-step)

1. Locate the branch collar
 2. Position the cutting tool slightly outside the collar
 3. Cut at a slight outward angle
 4. Complete the cut in one smooth motion
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Lesson 5.2 – Angle and Direction of Cuts

Proper Cutting Angle

- Always cut at approximately 45°
 - Angle away from the bud or trunk
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Purpose

- Prevents water from collecting on the cut surface
 - Encourages proper healing
 - Supports healthy new growth
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Example

Cutting a small citrus branch:

- Correct angle → water runs off → faster healing
 - Flat cut → water collects → risk of rot
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Demo

1. Identify the bud or branch collar
 2. Angle the tool away from it
 3. Make a smooth, angled cut
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Lesson 5.3 – Three-Cut Method for Large Branches

Definition

A technique used to safely remove large or heavy branches without damaging the tree.

Purpose

- Prevents bark tearing
 - Reduces risk of injury
 - Ensures clean removal
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Steps

1. First cut (Undercut)
 - Make a small cut underneath the branch
2. Second cut (Top cut)
 - Cut from the top, slightly beyond the first cut
 - Allows the branch to fall safely

3. Third cut (Final cut)
 - Remove the remaining stub at the branch collar
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Example

Removing a thick branch from a mango tree:

- Without method → bark tears and damages trunk
 - With three-cut method → clean and controlled removal
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Lesson 5.4 – Preventing Damage During Cutting

Common Risks

- Bark tearing
 - Splitting branches
 - Crushing plant tissue
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Prevention Tips

- Use sharp tools
 - Support heavy branches while cutting
 - Apply proper cutting techniques
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Example

Cutting a heavy branch improperly:

- Branch falls suddenly → tears bark

Correct approach:

- Controlled cuts → no damage to trunk
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Lesson 5.5 – Recognizing Proper vs Improper Cuts

Proper Cuts

- Clean and smooth
 - Slightly angled
 - Positioned correctly near collar or bud
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Improper Cuts

- Jagged or rough
 - Too close or too far from trunk
 - Flat or poorly angled
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Example

Two cuts on the same branch:

- Proper cut → heals naturally
 - Improper cut → becomes entry point for disease
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Lesson 5.6 – Practice Activity

Activity: Cut Evaluation

Observe different cuts on a tree and identify:

- Proper cuts
 - Improper cuts
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Guide Questions

- Is the cut smooth or rough?
 - Is it near the branch collar?
 - Is the angle correct?
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Lesson 5.7 – Quick Review

Key Points

- Always cut at the branch collar
- Use a 45° angle for effective healing
- Apply the three-cut method for large branches
- Avoid damaging bark and trunk
- Ensure all cuts are clean and precise